MIL-C-51096B
30 SEPTEMBER 1964
SUPERSEDING
MIL-C-51096A (MU)
15 JULY 1963

MILITARY SPECIFICATION

CORE MATERIAL, C18

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a filter medium containing charcoal and fibers.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

UU-P-268 — Paper, Kraft, Untreated, Wrapping.

UU-P-271 — Paper, Wrapping, Waterproofed Kraft.

PPP-F-320 — Fiberboard; Sheet, Stock and Cut Shapes.

PPP-P-291 — Paperboard, Wrapping, Cushioning.

PPP-T-76 — Tape, Pressure-Sensitive Adhesive Paper (For Carton Sealing).

PPP-T-97 — Taping, Pressure-Sensitive Adhesive, Filament Reinforced.

MILITARY

MIL-A-101 — Adhesive, Water-Resistant, For Sealing Fiberboard Boxes.

MIL-A-140 — Adhesive, Water-Resistant, Waterproof Barrier-Material.

MIL-C- 13724 — Charcoal, Activated, Impregnated, ASC.

MIL-M-50023 — Mask, Protective, Civilian, CD V-805.

MIL-F-50078 — Filter Element, Protective Mask, M13.

MIL-F-51095 — Filter Material, Gas-Aerosol, C18.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

FSC 4240

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:04Z
Check the source to verify that this is the current version before use.

MIL–STD–129 — Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM Standards

D645 — Thickness of Paper and Paper Products.

D646 — Basis Weight of Paper and Paper Products.

D828 — Tensile Breaking Strength of Paper and Paperboard.

(Copies may be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia 3, Pa.)

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Ratings, Rules and Regulations

(Application for copies of these freight classification ratings, rules, and regulations should be addressed to Uniform Classification Committee, 202 Union Station, Chicago, 6, Ill.)

3. REQUIREMENTS

3.1 Materials. The core material shall be manufactured from the following fibers and charcoal (see 6.4):

(a) Glass fibers, bulk: 0.5 to 0.75 micron fiber diameter.

(b) Vinyl chloride — vinyl acetate copolymer fibers: 85 to 88 percent vinyl chloride; 12 to 15 percent vinyl acetate; 3 denier.

(c) Regenerated cellulose fibers: 1½-denier.

(d) Impregnated charcoal conforming to grade I of MIL–C–13724, except that prior to aging it shall have a chloropicrin gas life of 28 minutes minimum and cyanogen chloride gas life of 40 minutes minimum. After acceptance, it shall be reduced in particle size to meet the requirements of table I.

TABLE I. *Particle size*

Distribution	Percent by weight
Passing #40 sieve	100
Retained on #50 sieve	0–2
Retained on #80 sieve	30–80
Retained on #100 sieve	10–40
Retained on #140 sieve	5–25
Retained on #200 sieve	0–10
Passing #200 sieve	0–7

3.2 Manufacture.

3.2.1 *Basis weight.* Prior to the addition of the charcoal at the start of any production, two consecutive fiber samples, taken five minutes apart, without adjustments to the equipment, shall have a basis weight of 49 ± 3 pounds per square feet (bone dry basis). After the fiber content basis weight has been established, the charcoal shall be added and the basis weight of two consecutive core material samples taken five min-

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:04Z
Check the source to verify that this is the current version before use.

utes apart, without adjustments to the equipment, shall be 220 ± 16 pounds per 3000 square feet (bone dry basis). The material on the reel shall be flagged to indicate the start of acceptable material.

3.2.2 *pH.* During production of core material, the pH in the head or collection box shall be maintained between 7.0 and 8.0. Dilute ammonium hydroxide shall be added if the pH is less than 7.0. The pH shall be continuously monitored by a recording pH meter.

3.3 **Physical properties.** The physical properties of the core material shall conform to table II.

3.4 **Size.** The core material shall be furnished in rolls with no more than 3 splices permitted for each roll and no piece less than 50 linear yards. The splices shall be lapped (approximately 1½ inches) and heat sealed across the width. The location of the splices shall be indicated by a paper flag of contrasting color. Width and diameter (including tolerances) of the roll shall be specified at the time of procurement (see 6.2).

3.5 **Preproduction sample.** At the discretion of the contracting officer prior to the start of regular production and following any change in regular production methods, materials or equipment, a preproduction sample of 12 rolls of core material shall be produced using the same methods, materials, and equipment as will be used during regular production. The preproduction sample shall pass the inspection specified in 4.3.

TABLE II. *Physical properties*

Property	Requirement	Test method
Airflow resistance, mm water.	(*)	4.7.1
DOP smoke penetration, %.	(*)	4.7.1
**Basis weight, bone dry, lbs/3000 sq. ft.	220 ± 16	4.7.2
Thickness, inches	0.053 ± 0.005	4.7.3
Average tensile strength, lbs/in. width machine direction cross direction.	 1.5 min 1.0 min	4.7.4
***Moisture content (immediately prior to packing).	5.0 max	4.7.5

TABLE II. *Physical properties* (Continued)

* The Government or contractor, whichever is applicable, shall specify the airflow resistance and DOP smoke penetration at the time of procurement. (See 6.2 and 6.7.)

** Not more than 15% of the rolls in any lot shall have a basis weight below 210 lbs/3000 sq. ft.

*** To prevent moisture pickup, the core material shall be packed for shipment within a half hour after being cut into rolls.

3.6 **Workmanship.** The core material shall be free from foreign matter, tears, punctures, pickouts, pull aparts, holes, thick or thin spots, burn holes, scorching, charring, wrinkles, pills or transparencies when examined as specified in 4.6.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 *Supplier's responsibility.* Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as other-

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:04Z
Check the source to verify that this is the current version before use.

wise-specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to insure that supplies and services conform to specified requirements.

4.2 Special provisions.

4.2.1 *Alternative inspection.* The supplier may utilize any alternative inspection procedure which will provide equal or better assurance of quality by submitting a written proposal with justification and obtaining written approval from the Government prior to instituting the procedure. In case of dispute, the procedures of this specification shall govern (see 6.6).

4.2.2 *Objective evidence.* The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1, 3.2, 3.4, and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2.3 *Substation inspection.* If the supplier elects to establish inspection substations for the purpose of conducting inspection for certain listed defects, the Acceptable Quality Levels (AQL's) for each inspection substation shall be equal to the AQL which is the next numerically lower value than the AQL specified for the total listed defects. The requirements of MIL–STD–105 are applicable to each substation and to the end item.

4.3 Preproduction sample inspection.

4.3.1 *Examination.* The preproduction sample shall be examined for all the defects in table III.

4.3.2 *Tests.* The preproduction sample shall be tested in accordance with table IV. Type II or type III filter material as specified (see 6.2) shall be fabricated and tested for conformance with MIL–F–51095. In addition, core material intended for use in the CDV–805 or the M17 mask shall be tested as follows:

4.3.2.1 *Type II filter material.* Fabricate 20 facepieces for the CDV–805 civilian protective mask and test for compliance to resistance to airflow and DOP smoke penetration requirements before and after rough handling specified in MIL–M–50023.

4.3.2.2 *Type III filter material.* Fabricate 20 matched pairs of filter elements for the M17 field protective mask and test for compliance to plastisol edgeseal, DOP smoke penetration, airflow resistance and gas life requirements specified in MIL–F–50078.

TABLE III. *Examination*

Defects	Number of defects per 10 sq yds		Action per roll
	1st sample	1st and 2nd sample combined	
Tears, punctures, pickouts, pull aparts or holes (exceeding 1/8 inch).	1	2	Accept
	2		Re-examine
	3 or more	3	Reject
Thick or thin spots	1	2	Accept
	2		Re-examine
	3 or more	3	Reject

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:04Z
Check the source to verify that this is the current version before use.

TABLE III. *Examination* (Continued)

Defects	Number of defects per 10 sq yds		Action per roll
	1st sample	1st and 2nd sample combined	
Transparencies (maximum dimension at least 1/4 inch).	5	10	Accept
	6–8		Re-examine
	9	11	Reject
Burn holes, charring scorching (exceeding 1/8 inch).	0 (max)		Accept
Wrinkles	0 (max)		Accept
Pills	10/sq foot (max)		Accept
Foreign matter, exceeding an area 1/4 x 1/4 inch.	0 (max)		Accept

TABLE IV. *Tests*

Test	Test paragraph	Test specimens at each end of roll*	Defective specimens at same end	Action per rolls	Retest specimens at end that fails	Defective specimens (test and retest combined)	Action per rolls
Airflow resistance and DOP smoke penetration.	4.7.1	3	0 1 2	Accept Retest Reject	3	1 2	Accept Reject
Basis weight.	4.7.2	2	0 1 2	Accept Retest Reject	2	1 2	Accept Reject
Thickness	4.7.3	3	2	Accept Retest Reject	3	1 2	Accept Reject
Tensile strength.	4.7.4	3 in each direction.	Reject the roll if the average in either direction does not meet the tensile strength requirement.				
Moisture content.	4.7.5	1	0 1	Accept Retest	2	1 2	Accept Reject

* If the end of one roll is adjacent to the beginning of the following roll, specimens need be taken only at the open end and the results used for both rolls.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:04Z
Check the source to verify that this is the current version before use.

4.3.3 *Acceptance/rejection criteria.* The preproduction sample shall meet the requirements of 4.3.1 and 4.3.2 to be acceptable.

4.4 Lotting. A lot shall consist of the core material produced by one manufacturer from no more than one lot of charcoal under essentially the same manufacturing conditions without any change in materials. A new lot shall be established after a shutdown exceeding eight hours.

4.5 Sampling.

4.5.1 *For examination.*

4.5.1.1 *Rolls.* A six-foot long sample, full width, shall be taken from each roll. Each sample shall be identified with the roll represented.

4.5.1.2 *Preparation for delivery.* Samples shall be selected in accordance with MIL–STD–105.

4.5.2 *For destructive tests.* An appropriate size strip shall be taken across the width of each reel. Each strip shall be divided into samples which shall be identified with the roll represented.

4.6 Inspection Provisions.

4.6.1 *Rolls.* Prior to reeling, a visual examination of the core material shall be conducted over a source of light under the full width of the reel. Ten square yards representing each component roll shall be examined for defects in accordance with table III. Re-examination shall be made on an additional ten square yards if permitted by table III. In addition each 6-foot sample taken as specified in 4.5.1 shall be examined over a light box in accordance with table III, except that re-examination shall not be allowed.

4.6.2 *Preparation for delivery.* Sample rolls shall be examined for level B requirements in accordance with the classification of defects and with MIL–STD–105.

4.6.3 *Classification of defects.*

4.6.3.1 *Preparation for delivery.*

Categories	*Defects*
Critical:	None defined.
Major:	AQL 2.5 per cent defective.
101	Spirally wound core not as specified.
102	Initial, intermediate or final wrap not as specified.
103	Circular pads not as specified.
104	Circular pads missing or incorrectly placed.
105	Sealing material missing or inadequate.
106	Marking missing, incorrect, or illegible.

4.7 Tests. Each roll sample taken as specified in 4.5.2 shall be tested in accordance with table IV. Test specimens shall be taken at random from each sample.

4.7.1 *Airflow resistance and DOP smoke penetration.* The test specimens (5¼ inch diameter discs) shall be tested for airflow resistance and DOP smoke penetration at a flow rate of 32 liters per minute, using the Edgewood Arsenal Q127 DOP Filter Testing Penetrometer.

4.7.2 *Basis weight.* The basis weight shall be determined in accordance with ASTM D646 except that conditioning of specimens shall be omitted. The ASTM basis weight shall be corrected to a bone dry basis as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:04Z
Check the source to verify that this is the current version before use.

$$\text{Bone dry basis weight} = \text{ASTM basis weight} \frac{(100 - \%\ \text{moisture})}{100}$$

where percent moisture is determined as specified in 4.7.5.

4.7.3 *Thickness.* Thickness of the test specimens shall be determined in accordance with method C of ASTM D645 except that conditioning of specimens shall be omitted. Three determinations shall be made on each specimen and the average reported as the thickness.

4.74 *Average tensile strength.* The tensile strength of the test specimens shall be determined in accordance with ASTM method D828 except that conditioning of specimens shall be omitted.

4.7.5 *Moisture content.* Each test specimen (5¼-inch diameter discs) shall be weighed to the nearest 0.01 gram (wet weight). Place the specimens in a forced air drying oven at 105° to 110°C. until constant dry weight is reached. Record the final weighting (dry weight) to the nearest 0.01 gram. Calculate the percent moisture as follows:

$$\text{Percent moisture content} = 100\ \frac{(\text{Wet weight} - \text{dry weight})}{\text{wet weight}}$$

5. PREPARATION FOR DELIVERY

5.1 Packing. Level shall be as specified (see 6.2).

5.1.1 *Level B.* The material shall be wound evenly on a spirally wound minimum ⅜-inch thick wall hollow core constructed of fiberboard or other suitable material having an inside core diameter of 3 ± ⅛-inch. Individual rolls of material shall have an initial wrap of Grade B, 40-pound basis weight Kraft wrapping paper conforming to UU–P–268, secured in place with minimum two-inch wide pressure-sensitive tape conforming to PPP–T–76. Each roll shall then have an intermediate wrap, consisting of a minimum of three complete wraps of type III, class 2, paperboard, conforming to PPP–P–291, secured in place with the tape cited above. Four circular pads, fabricated from type CF, class weather-resistant, variety SW, grade V3c, fiberboard, conforming to PPP–F–320, laminated together with water-resistant adhesive, conforming to MIL–A–101, shall be placed at each end of the wrapped roll. The diameter of the circular fiberboard pads shall be the same as the overall diameter of the intermediate wrapped rolls. Each circular fiberboard pad shall be secured in place with a minimum of ten strips of 2-inch wide by 12-inch long pressure-sensitive tape, conforming to type IV of PPP–T–97, equally spaced around the circumference of the fiberboard pad. Each roll shall then be overwrapped with class E-3 or M-3 waterproofed Kraft wrapping paper conforming to UU–P–271, sealing all folds, laps, and seams in place with water-resistant adhesive conforming to MIL–A–140.

5.1.2 *Level C.* The material shall be packed in a waterproof pack in such manner as to insure carrier acceptance and safe delivery at first domestic destination. Containers shall be in accordance with Uniform Freight Classification rules and regulations of other carriers applicable to the mode of transportation.

5.2 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with MIL–STD–129.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:04Z
Check the source to verify that this is the current version before use.

5.2.1 *Special marking.* In addition to the specified identification markings, the special markings on the roll shall be as follows:

(a) Date of manufacture.

(b) Lot number of charcoal.

(c) Average bone dry basis weight by ASTM method.

(d) Store in cool and dry location.

5.2.2 *Fragile marking.* The following marking shall be minimum 1-inch high letters, and shall be applied in two places on the body of the roll or container in such locations so as not to interfere with other specified markings:

FRAGILE, HANDLE WITH CARE,

DO NOT DROP

6. NOTES

6.1 Intended use. The material covered by this specification is intended for use in the manufacture of C18 Gas-Aerosol Filter Material.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Level of packing required.

(c) Roll width and diameter including tolerances.

(d) Type and use of filter material (see 4.3.2).

(e) Percent DOP penetration and airflow resistance (see 6.7).

(f) Laboratory to which samples are to be forwarded.

6.3 Significant places. For the purpose of determining conformance with this specification, an observed or a calculated value shall be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of the Recommended Practices for Designating Significant Places in Specified Limiting Values (ASTM Designation: E29).

6.4 Composition. The following composition has been found to be satisfactory:

AAA 475–106 J–M Glass Fibers	1.5 percent by weight.
Vinyon HH (1/8 or 1/4-inch staple, 3 denier)	10.4 percent by weight.
Viscose rayon 1/4-inch maximum staple, 1½ denier)	10.4 percent by weight.
Charcoal (Fine particle Whetlerite)	77.7 percent by weight.

6.5 Precaution. For retention of maximum gas life, core material should be used prior to six months from the date of manufacture for the fabrication of gas-aerosol filter material.

6.6 For nonmilitary procurement, the procuring agency shall be responsible for preproduction sample inspection and for approval of alternative inspection procedures.

6.7 Core and backing material. Core and backing materials having the following phyical properties are recommended for the manufacture of type II or III, C18 Gas-Aerosol Filter Materiel covered by MIL–F–51095:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:04Z
Check the source to verify that this is the current version before use.

Percent of each shipment	Core material		Backing material			
	For type II or III		For type II		For type III	
	Percent DOP	Air resistance	Percent DOP	Air resistance	Percent DOP	Air resistance
85 min	70 max	2.7 max	5 max	13 max	8 max	12 max
15 max	80 max	3.0 max	8 max	14 max	10 max	13 max

6.8 International interest. Certain provisions of this specification are the subject of ABC international standardization agreement. When amendment, revision, or cancellation of this specification is proposed, the departmental custodians will inform their respective Departmental Standardization Offices so that appropriate action may be taken respecting the international agreement concerned.

Custodians:

Army — MU

Navy — SH

Air Force — 84

International interest (see section 6)

Review activities:

Army — MD

Navy — SH

User activities:

Army — MU

Navy — MC

Air Force — 84

Preparing activity:

Army — MU (EA)

Project No. 4240–0110

☆ U. S. GOVERNMENT PRINTING OFFICE: 751072—1965—(T-150)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:04Z
Check the source to verify that this is the current version before use.